BockNet: Blind-Block Reconstruction Network With a Guard Window for Hyperspectral Anomaly Detection

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***Abstract*— Hyperspectral anomaly detection (HAD) aims to identify anomalous targets that deviate from the surrounding background in unlabeled hyperspectral images (HSIs). Most existing deep networks that exploit reconstruction errors to detect anomalies are prone to fit anomalous pixels, thus yielding small reconstruction errors for anomalies, which is not favorable for separating targets from HSIs. To achieve a superior background reconstruction network for HAD purposes, this article proposes a self-supervised blind-block network (termed BockNet) with a guard window. BockNet creates a blind-block (guard window) in the center of the network's receptive field, rendering it unable to see the information inside the guard window when reconstructing the central pixel. This process seamlessly embeds a sliding dual-window model into our BockNet, in which the inner window is the guard window and the outer window is the receptive field outside the guard window. Naturally, BockNet utilizes only the outer window information to predict/reconstruct the central pixel of the perceptive field. During the reconstruction of pixels inside anomalous targets of varying sizes, the targets typically fall into the guard window, weakening the contribution of anomalies to the reconstruction results so that those reconstructed pixels converge to the background distribution of the outer window area. Accordingly, the reconstructed HSI can be deemed as a pure background HSI, and the reconstruction error of anomalous pixels will be further enlarged, thus improving the discrimination ability of the BockNet model for anomalies. Extensive experiments on four datasets illustrate the competitive and satisfactory performance of our BockNet compared to other state-of-the-art detectors.**

***Index Terms*— Blind-spot network, convolutional neural network, deep learning (DL), hyperspectral image (HSI), image reconstruction, neural network, self-supervised learning.**

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I. Introduction

HYPERSPECTRAL remote-sensing images depict both the spectral and spatial characteristics of ground objects [1], [2] and this detailed information provides valuable data for identifying ground materials [3], [4]. As an important part of the hyperspectral image (HSI) interpretation task, hyperspectral anomaly detection (HAD) aims to locate targets that deviate from the surrounding background distribution from a spectral and spatial perspective. These targets are typically referred to as anomalies [5], [6]. The difficulty of HAD lies in separating anomalies and background from HSIs without any prior knowledge [7], [8]. Researchers have explored new techniques to model the background and identify abnormal properties from multiple aspects. Existing HAD methods can be roughly classified into three categories, namely statistical modeling-based, representation learning-based, and deep-learning (DL)-based techniques.

A. Statistical Modeling-Based Methods

The Reed-Xiaoli (RX) detector is a classic and widely used statistical-based method, which assumes that the background data in HSIs obeys a multivariate Gaussian distribution and generally models the background using global information of the HSI [9]. To reduce the interference of anomalous pixels and further improve the detection effect of the model in complicated scenes, many RX-based derivatives have been proposed. For example, the local RX (LRX) [10] method constructs statistics only from the surrounding pixels of the testing pixel (instead of the global pixels). Then, the segmented RX (SRX) [11] uses the neighborhoods of each pixel on the k-means clustered HSI to detect anomalies. In addition, Tao et al. [12] transformed all pixels to an intermediate domain, for example, fractional Fourier domain (FrFD), before performing RX detection. This enhanced the separability of anomalies from the background. Liu et al. [13] designed a two-step GLRT method using a sliding dual-window strategy, called 2S-GLRT, for the case of multipixel anomalies with unknown patterns. Different from the above methods, Vincent et al. [14] modified the RX scheme and adopted

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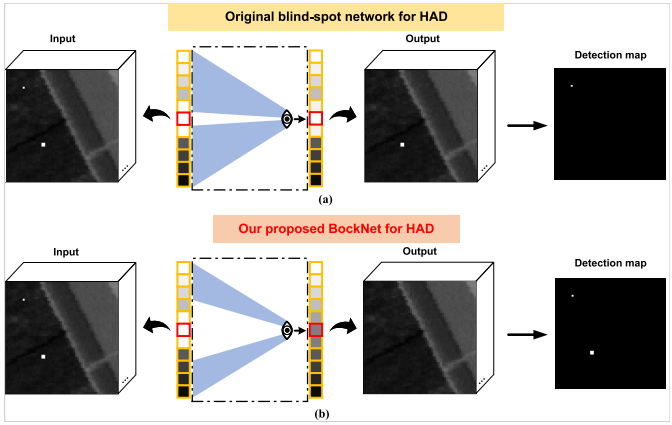


Fig. 1. Blind-spot image reconstruction network is suitable for the HAD task because its receptive field has a blind-spot in the center (meaning that the center pixel is invisible to the network/eye). As a result, the network has to predict/reconstruct the center pixel using only the surrounding pixels (blue areas seen by the eye). This causes the network to generate high reconstruction errors for anomalous pixels and low reconstruction errors for background pixels. (a) Original blind-spot network has a blind area in the receptive field corresponding to a single central pixel, which is advantageous in detecting small-sized targets. However, anomalous targets with a relatively larger spatial size are likely to be reconstructed well because the reconstruction of anomalous pixels will be interfered with by the adjacent pixels that are also anomalies, which eventually leads to undesirable results. (b) Our proposed BockNet improves upon the original blind-spot network by using a central block (i.e., the white area invisible to the eye) as the blind area in the receptive field. This central block provides a guard window for reconstructing the central pixel and avoids the potential contamination of anomalous pixels. As a result, BockNet can show excellent detection performance for both large- and small-sized targets, extending the application scope of the blind-spot model.

a replacement model as a more realistic framework for anomaly detection. To fully exploit the multiple features in HSI, Yang et al. [15] fused the detection results of different features on the random RX detector aided by ensemble learning. Besides, diverse techniques were applied to HAD, such as SVDD using the kernel trick avoids a priori assumption [16], covariance analysis with similar specifications has been employed to extract both spatial and spectral characteristics of HSIs [17], [18], [19], and Parzen Windowing-based detector [20] enables nonparametric background distribution estimation to detect global anomalies in an image.

B. Representation Learning-Based Methods

Later, representation learning-based methods, which incorporate sparse representation (SR) [21], [22], [23], collaborative representation (CR) [24], and low-rank representation (LRR) [25], [26], [27], have received extensive attention and discussion among researchers [28], [29]. The core idea of SR-based approaches is that normal samples can be represented by a small number of atoms in an overcomplete background dictionary, while anomalous ones cannot [30]. The CR-based detector (CRD) [31] places more emphasis on the collaboration relationship of all dictionary atoms, aiming to focus on whether each pixel can be linearly represented by the surrounding pixels. Both types of methods treat the reconstruction residuals as anomalous representations of the testing pixels. To obtain accurate detection results, Vafadar and Ghassemian [32] improved CRD by removing outliers from the image before performing background estimation. To lessen the contamination of abnormal data when constructing background dictionaries, Hou et al. [33] proposed a background purification strategy to remove the impact of potential outlier pixels, which has become an effective CRD-based variant. The global structure of an HSI is viewed from the perspective of LRR, with the low-rank properties of backgrounds and the sparse properties of anomalies taken into account [34], [35]. Zhang et al. [36] utilized the Mahalanobis distance between each testing pixel and the extracted low-rank matrix for HAD. Furthermore, Li et al. [37] adopted a model with a mixture of Gaussian noise to represent the sparse component, which detects anomalies by using the Manhattan distance. Besides, LRR-based detectors can also construct the background dictionary using original data to better reconstruct HSIs and highlight anomalies. Meanwhile, a variety of background dictionary construction methods emerged. Xu et al. [38] and Fu et al. [39] chose to perform k-means clustering on HSIs to obtain background dictionaries, while Qu et al. [40] employed the mean-shift clustering approach in the abundance matrix to construct the dictionary, and separated anomalous samples from it, which weakened the influence of noise. Taking into account the correlation between neighboring pixels, the total variation technique is introduced into LRR [41] to make the atoms in the dictionary represent the background more effectively.

C. DL-Based Methods

With its powerful ability to interpret high-dimensional and massive data, DL has emerged as an efficient tool in many tasks [42], [43], [44]. The mainstream DL-based HAD methods can be further subdivided into similarity measurement-based and reconstruction model-based detectors.

Similarity measurement-based methods require a large number of training sample pairs to be selected on the reference classification HSI with labels. If the paired samples come from the same category of ground objects, they are labeled as 0 and 1 otherwise. Li et al. [45] trained a convolution neural network (CNN) with these pixel pairs and transferred it to the HAD task. The average difference score between each pixel to be measured and the outer window pixels is used to represent the degree of anomaly of the corresponding central pixel. Analogously, Rao et al. [46] designed a Siamese network architecture with feature extraction and similarity measurement functions and identified those pixels that are significantly different from the surrounding samples at the hidden level. However, the performance of these metric networks is highly dependent on the quantity of paired data and the quality of annotations, and the transferability will be relatively limited. The whole process is costly in terms of time and labor.

Conversely, reconstruction model-based approaches are more compatible with the unsupervised HAD task, as they directly employ all samples in the unlabeled HSI as training data to mine the in-depth, intrinsic spectral characteristics in an unsupervised learning (USL) manner. These typical methods mainly rely on autoencoders (AEs) and generative

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adversarial networks (GANs) and adopt reconstruction errors to reflect the abnormal level of each pixel. Bati et al. [47] introduced AEs into HAD for the first time in the literature and attempted to reconstruct all pixels in an HSI. Since background samples account for the vast majority of the original HSI, the model can learn the background representations adequately and reconstruct the background excellently. On the contrary, the proportion of abnormal samples is relatively low, and they exhibit large reconstruction errors. To explore the spatial characteristics of HSIs, Fan et al. [48] introduced graph regularization in the hidden layer of AEs (RGAEs) to improve the robustness of the model, while Xiang et al. [49] imposed a guided module constraint on the latent features of AEs (GAEDs) to enhance the model's ability to represent backgrounds. Then, Wang et al. [50] built a fully convolutional AE to perform automatic end-to-end HAD (Auto-AD). Additionally, unlike traditional AEs, the memory AE, with an additional memory-augmented module to record the latent representation of background samples, was introduced to the HAD task [51], [52]. To motivate AE networks to characterize the background exhaustively and reconstruct it adequately, the discriminator is embedded in the AE (constituting GAN) by imposing the Gaussian distribution constraint on it [53] or a rough background sample screening process is designed before training the model [54], [55], accordingly increasing the reconstruction error of anomalies. More recently, Li et al. [56] chose a parameter-free mean-shift clustering-based strategy to select relatively pure background pixels and further integrated spectral and spatial features by a two-stream convolution AE network architecture.

D. Motivation and Contributions

Among the various HSI anomaly detection approaches mentioned above, similarity measurement-based detectors require complex procedures for generating pixel pairs for network training, and the model generalization is limited in practical applications. On the other hand, most reconstruction model-based detectors that take the input data as a target tend to capture low-dimensional features between background and anomalies and learn an identity mapping [57], [58], thus making it difficult to distinguish anomalies. Anomalies are often characterized by small reconstruction errors similar to backgrounds, which increases the difficulty of identifying anomalous pixels based on such reconstruction errors.

To address this issue, this article adopts a self-supervised learning (self-SL) technique for HAD. Self-SL is similar to USL in that no labels are given. However, self-SL generates supervisory signals from the input data and then uses them for training. Self-SL comprises two key steps: 1) constructing input samples and corresponding supervision information on unlabeled data and 2) designing the supporting pretext task according to specific target tasks, aiming to extract the intrinsic data features that are profitable to the target task [59]. Currently, self-SL techniques have been successfully applied in various HSI interpretation tasks, such as image restoration [60], [61], [62], unmixing [63], classification [64], and so on, with promising results.

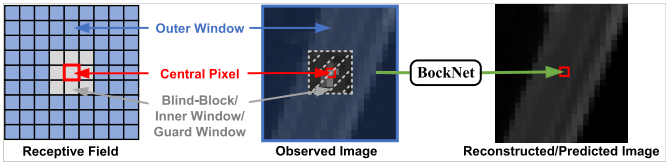


Fig. 2. Illustration of the dual-window provided by our BockNet. The left, middle, and right images denote the receptive field, observed, and predicted images of the network. To reconstruct the central pixel (represented by the red rectangle), BockNet imposes a blind-block with the gray rectangle (which is invisible to the network) in the center of the receptive field (represented by the blue rectangle), enabling the original receptive field to be divided into a dual-window, where the blind-block is viewed as the inner window (guard window), and the receptive field outside the blind-block (represented by the blue area) is treated as the outer window. Hence, only the information on the outer window is employed to predict the central pixel.

As one of the most prominent self-SL strategies, the blind-spot network possesses a special receptive field characterized by a central blind-spot, meaning the central pixel is invisible to the network. Consequently, the network's prediction for a pixel is influenced by all surrounding pixels within the receptive field, excluding the central one [65]. It has recently been introduced into HAD [66] showing excellent detection performance when identifying small-sized subpixel anomalous targets. However, this technique does not work effectively with large-sized targets. A toy example is shown in Fig. 1 to illustrate the challenge of detecting large-size targets using the blind-spot network. We emphasize that the blind-spot network cannot see the single-pixel information in the center (blind-spot) of the receptive field, but only predicts the blind-spot using its neighboring pixels [67], [68]. When the central pixel is an anomaly with a significant difference in spectral shape from its neighbors, the blind-spot network is unlikely to reconstruct the center pixel perfectly, leading to an increase in the reconstruction error of the anomaly. Therefore, the architecture of blind-spot networks naturally suits the HAD task quite well. However, as shown in Fig. 1(a), we can observe that the original blind-spot network is not effective for detecting large-sized targets. It happens that the network can predict the center anomalous pixel from its neighboring pixels very well because an anomalous target is large in size and the neighboring pixels contain spectral information of the target.

To cope with this challenge, we propose a new blind-block network with a guard window for HAD. As illustrated in Fig. 1(b), our network does not see the information of the center block (blind-block) of the receptive field during the process of pixel reconstruction but utilizes the pixels outside the blind-block to predict the corresponding central pixel. This procedure can evolve into a sliding dual-window-based detector, such as LRX, which only applies the outer window pixels to model the background, while the inner window serves as a guard window against potential contamination caused by anomalous pixels. Our network divides the original receptive field into a dual-window, where the inner window (guard window) is the blind-block and the outer window is the receptive field outside the blind-block. Therefore, only the receptive field of the outer window is employed to reconstruct the central pixel. The illustration is shown In Fig. 2. When

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reconstructing central anomalous samples, the sparsely distributed and low-proportion target with a certain size typically lies in the inner window, and the network does not see these anomalous properties. Moreover, the background is generally a large-scale homogeneous region, which tends to be located within the outer window, and then the samples reconstructed by the network are more probable to converge to the surrounding background distribution of the outer window, hence increasing the reconstruction error of the anomaly. Specifically, we build a network structure in which the receptive field grows along four different directions (up, left, down, and right), and the receptive field is in a half-plane containing the central pixel. Among them, a simplified version of U-Net (including an encoder and a decoder) is designed to fully explore the spatial–spectral characteristics of HSIs. In addition, to set up the guard window, the four feature maps output by the decoder require to be shifted and combined by the fusion module to obtain a receptive field that can be extended arbitrarily in each direction but does not contain the central blind-block. This greatly reduces the contamination of anomalous pixels to the receptive field applied to reconstruct the central pixel, thereby weakening the ability to fit anomalous characteristics. Meanwhile, the whole network can be considered as a superb background reconstruction model. Finally, the degree of anomaly of each pixel in HSIs is represented by the corresponding reconstruction error generated by our network. Compared to existing HAD methods, the main contributions of our proposed **B**lind-bl**ock** reconstruction **Net**work with a guard window (BockNet) are fourfold.

1) We propose a new blind-block network (BockNet) for HAD in a fully self-SL scheme, requiring no labeled samples for network training, yet achieving promising detection results compared to the existing unsupervised HAD networks.
2) The use of blind-block architecture in our BockNet is well-suited to the HAD task. Pixels within the central block of the perceptive field are invisible to our network, and the network has to predict/reconstruct the central pixel using the remaining surrounding pixels. As anomalous targets significantly differ from surrounding background pixels, our network exhibits large reconstruction errors, which can be used to distinguish anomalies from background pixels.
3) This study endows the blind-spot learning architecture with the ability to detect anomalous targets of slightly larger sizes by introducing a guard window. Specifically, our network works as a sliding dual-window reconstruction model, in which the inner window is the guard window (blind-block), and the outer window is the receptive field outside the guard window. The use of the guard window prevents large-size anomalies (lying in the inner window) from contributing to the reconstruction of the central pixel.
4) We implemented blind-blocks of varying sizes in BockNet by employing convolution and pooling operations with an offset process to acquire a receptive field growing along four directions (up, left, down, and right) within the half-plane containing the central pixel, as well as performing shifting with different strides and fusing operations on the feature maps. Therefore, our BockNet is flexible for identifying both large- and small-size targets, extending the application range of the series of blind-spot models.

This work is an expanded version of our conference paper [69], with the added material including mainly more detailed method descriptions and experimental analysis.

The remainder of the article is organized as follows. Section II describes the details of the proposed BockNet. In Section III, extensive experimental results and a comparison analysis between BockNet and state-of-the-art methods are presented. Finally, our conclusion is drawn in Section IV.

II. Proposed Method: BockNet

A. Overview

With the cognition of the properties of anomalous targets in an HSI on spatial and spectral aspects, we design the BockNet framework to be suitable for detecting targets of different sizes, and its overall flowchart is described in Fig. 3. More specifically, BockNet can be divided into two stages.

1) *Training Phase:* First, a simplified U-Net architecture with the receptive field extended in a specific direction is built. Then, different rotated versions of HSIs are fed into the network and operated by shifting. Finally, the shifted feature maps are rotated back and fused to produce the reconstructed HSI, where BockNet is achieved with a blind-block (the whole network is unable to see this information) in the center of the receptive field, and the blind-block acts as a guard window. The objective function is set as a reconstruction loss based on the L1 norm.
2) *Detection Phase:* This follows the same behavior as the data processing in the training phase, where the original HSI is fed into the trained model to gain a relatively pure background HSI and the reconstruction error is taken as the HAD result. Next, we introduce each part of our BockNet in detail.

B. Problem Formulation

Let us denote an unlabeled HSI as $\mathbf{X} \in \mathbb{R}^{R\times R\times L}$, where R and L, respectively, represent the row number (column number) and band number (here, the channel number) of $\mathbf{X}$. The proposed BockNet seeks to be a superior background image reconstruction model in a self-supervised scheme, that is, only using the observed HSI $\mathbf{X}$ without labeled background information. BockNet imposes a guard window for its own receptive field, which implies that the pixel information within the guard window does not require to be exploited when reconstructing the center pixel of the receptive field. In the reconstruction of anomalous samples, the entire target would generally be inside the guard window, which can decrease the contribution of anomalies. Meanwhile, the network only uses the receptive field outside the guard window (outer window) to reconstruct the center pixel, where generally the background of a large homogeneous region tends to be

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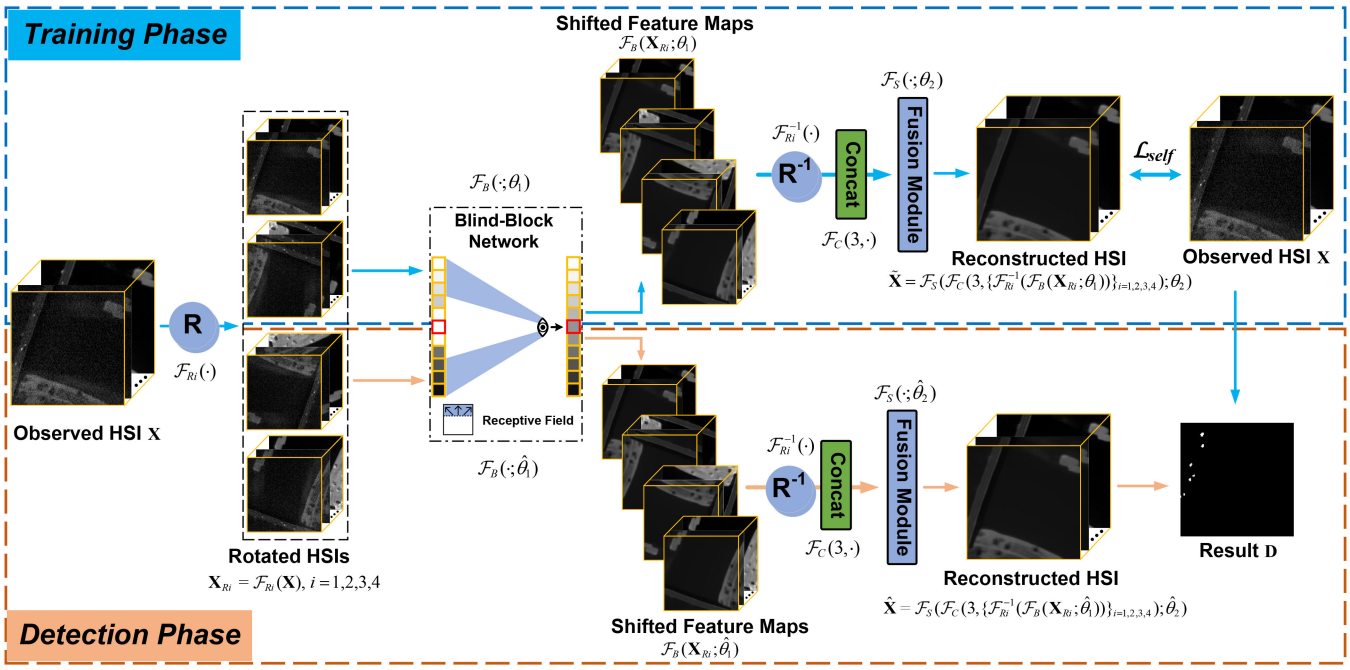


Fig. 3. Flowchart of the proposed BockNet for HAD.

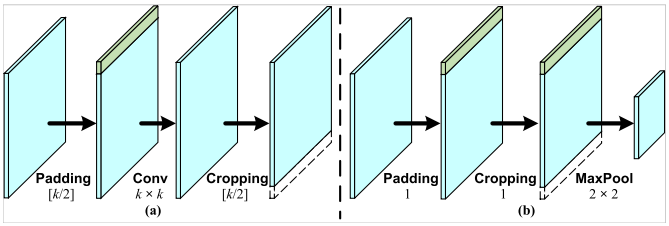


Fig. 4. Modified (a) convolution layer (offset) and (b) max-pooling layer as the downsampling layer (offset).

located inside the outer window, rendering the background samples sufficiently reconstructed. Consequently, our network boosts the feature representation of the background, yielding a low reconstruction error for background samples and a high reconstruction error for anomalous samples. Naturally, the reconstruction errors indicate abnormal levels for each pixel and help recognize anomalous targets from the original HSI.

C. Training Phase

This section mainly introduces the specific architecture of BockNet and its training phase.

First, we reiterate that the core concept of blind-spot networks is that the network predicts the invisible central pixel in its receptive field using its neighboring pixels. This mechanism is appropriate for removing the anomalous properties of small-sized targets in the image to converge to the distribution of contextual backgrounds, but is unsuitable for HSIs containing large-sized targets, because of the interaction among the anomalous pixels. Our BockNet, in contrast, reconstructs one pixel using its neighboring pixels and ignoring pixels lying in the central block of its receptive field (meaning that our blind area is larger than that of the original blind-spot network). Note that our blind-block is essentially serves as a guard window. When our network reconstructs anomalous pixels, targets of a particular size usually fall into the guard window. The information reconstructed by the network is expected to satisfy the background distribution outside the guard window, mitigating the contamination of anomalous pixels. Therefore, the guard window endows our BockNet with the ability to detect targets with different sizes in the image. Actually, both the blind-spot network and the blind-block network are constructed on the principle of mutual coherence, that is, the blind area (blind-spot or blind-block) is directly created in the center of the receptive field. Here, we adopted a network in which the receptive field grows only upward along the half-plane and feeds the original HSI into this network to achieve a receptive field that extends in all directions without the center area (spot or block) through operations like rotation, shift, and fusion.

Concretely, the convolution and pooling layers were modified (see Fig. 4) to achieve the functionality that the receptive field be restricted to the upper half-plane containing the central row. For the convolution layer with kernel size $k \times k$, the input feature map is offset downward by $d = [k/2]$ pixels in advance, with the specific implementation described in Fig. 4(a), where the top of the feature map is first padded with d rows of zeros, then the traditional convolution is applied, and finally the last d rows of the feature map are cropped. This behavior is equivalent to using a convolution kernel with a parameter of 0 below the center row to restrict its receptive field to grow upward only, as shown in Fig. 5(a). For the 2×2 max-pooling layer, which is taken as a downsampling step and paired with an upsampling step based on nearest neighbors. If these layers are left unconstrained, the receptive field of the network will grow in every direction. Consequently, the input feature map is offset down by 1 pixel before pooling, and the detailed procedure is presented in Fig. 4(b) which starts

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by adding 1 line of zeros to the top of the feature map and cropping the last line of the feature map before performing the pooling operation. The upsampling step does not require to be modified exceptionally. In fact, operations such as skip connection, 1×1 convolution, summation, and concatenation will not extend the receptive field and thus can be applied without modification.

With the analysis above, a simplified U-Net (including an encoder and a decoder) is built to exploit the spatial–spectral features of the HSI. Its receptive field can only grow upward along the center row. The detailed network structure is described in Fig. 6, where the encoder mainly consists of one modified max-pooling layer as the downsampling layer (offset) and five modified 3×3 convolution layers (offset) immediately following the Leaky rectified linear unit (Leaky ReLU) layer, and similarly, the decoder is composed of one nearest neighbor-based upsampling layer and seven modified 3×3 convolutional layers (offset) that immediately follow the Leaky ReLU layer. Additionally, there are different layers of feature maps for concatenation, to integrate the global and local detail characteristics in the HSI. Before the observed HSI $\mathbf{X}$ is fed into the simplified U-Net, we rotate the image in four directions as follows:

$$\mathbf{X}_{Ri} = \mathcal{F}_{Ri}(\mathbf{X}), \quad i = 1, 2, 3, 4 \tag{1}$$

where $\mathcal{F}_{Ri}(\cdot)$ refers to the ith rotation operation corresponding to four rotation angles: 0°, 90°, 180°, and 270°, respectively, and $\mathbf{X}_{Ri} \in \mathbb{R}^{R \times R \times L}$ indicates the ith rotated HSI. If the four rotated HSIs $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$ are fed into the simplified U-Net (parameterized by θ_1) and directly rotated back to the correct directions, the actual receptive fields for the center pixels of the feature maps produced by the decoder are presented in Fig. 5(b) (the number of channels is omitted for simplicity). These four receptive fields extend along four directions and encompass the central row/column. Next, we perform a shift operation on the output feature maps of the simplified U-Net to yield the four restricted receptive fields on the left part of Fig. 5(c) if the shift stride s is 1, and the four restricted receptive fields on the left part of Fig. 5(d) if the stride s is 2. These receptive fields will not consist of central pixels. Here, we denote the network including the simplified U-Net and shift operation as $\mathcal{F}_B(\cdot; \theta_1)$, so that the output corresponding to the input $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$ is the four shifted feature maps, denoted as $\mathcal{F}_B(\mathbf{X}_{Ri}; \theta_1)$.

Finally, we rotate $\mathcal{F}_B(\mathbf{X}_{Ri}; \theta_1)$ back to the corresponding initial direction, concatenate them, and apply the fusion module $\mathcal{F}_S(\cdot; \theta_2)$ (which is composed of 1×1 convolution) to acquire the final reconstructed image $\widetilde{\mathbf{X}} \in \mathbb{R}^{R \times R \times L}$, that is,

$$\widetilde{\mathbf{X}} = \mathcal{F}_S\left(\mathcal{F}_C\left(3, \left\{\mathcal{F}_{Ri}^{-1}(\mathcal{F}_B(\mathbf{X}_{Ri}; \theta_1))\right\}_{i=1,2,3,4}\right); \theta_2\right) \tag{2}$$

where $\mathcal{F}_C(3, \cdot)$ is the concatenation operation along dimension 3, and $\mathcal{F}_{Ri}^{-1}(\cdot)$ is the inverse operation of $\mathcal{F}_{Ri}(\cdot)$.

By doing this, we obtain the receptive field of our BockNet, which can be extended arbitrarily in each direction but does not embrace the central blind area (blind-spot or blind-block). The diagrams on the right part of Fig. 5(c) and (d) show final receptive fields while the stride (denoted as s) is set to 1 and 2, respectively. When the shift stride s is 1, the final receptive field has a blind-spot at the center pixel position, and the whole network can be treated as a blind-spot network without the guard window. When the stride s is 2, the final receptive field contains a blind-block at the center, which is viewed as a guard window of size $B_s \times B_s$ (where window width $B_s = 2 \times s - 1$). As it can be observed from Fig. 5(d), the original receptive field is divided into a dual-window, with the inner window being the guard window and the outer window being the receptive field outside the guard window. Only the outer window pixels are available for reconstructing the central pixel. This indicates that each pixel reconstructed by our BockNet (coupled with the sliding dual-window model) will satisfy the background distribution within the outer window, weakening the network's capability to express anomalous samples.

We did not adopt the typical ℓ_2 loss given that it would be more concerned with larger errors but less sensitive to small ones [70]. For better generalization, our BockNet $\mathcal{F}(\cdot; \theta)$ is trained using the following ℓ_1 loss function in the following equation:

$$\begin{aligned} \hat{\theta} &= \arg\min_{\theta} \mathcal{L}_{\text{self}} \\ &= \arg\min_{\theta} \|\mathcal{F}(\mathbf{X}; \theta) - \mathbf{X}\|_1 \end{aligned} \tag{3}$$

where

$$\begin{aligned} &\mathcal{F}(\mathbf{X}; \theta) \\ &= \mathcal{F}_S\left(\mathcal{F}_C\left(3, \left\{\mathcal{F}_{Ri}^{-1}(\mathcal{F}_B(\mathcal{F}_{Ri}(\mathbf{X}); \theta_1))\right\}_{i=1,2,3,4}\right); \theta_2\right) \end{aligned} \tag{4}$$

where $\mathcal{F}_C(3, \cdot)$, $\mathcal{F}_{Ri}(\cdot)$, and $\mathcal{F}_{Ri}^{-1}(\cdot)$ are concatenation, rotation, and inverse rotation operations, respectively. $\mathcal{F}_B(\cdot; \theta_1)$ denotes the network, parameterized by θ_1, including the simplified U-Net and shift operation, $\mathcal{F}_S(\cdot; \theta_2)$ denotes the fusion module, parameterized by θ_2.

D. Detection Phase

In the detection phase, we analogously generate four rotated HSIs $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$ from $\mathbf{X}$ first and then feed them into the well-trained BockNet, parameterized by $\hat{\theta}$ (composed of $\hat{\theta}_1$ and $\hat{\theta}_2$). The final reconstructed image (denoted as $\widehat{\mathbf{X}} \in \mathbb{R}^{R \times R \times L}$) is obtained through a series of operations, such as rotation, fusion, and so on, as follows:

$$\widehat{\mathbf{X}} = \mathcal{F}_S\left(\mathcal{F}_C\left(3, \left\{\mathcal{F}_{Ri}^{-1}\left(\mathcal{F}_B\left(\mathbf{X}_{Ri}; \hat{\theta}_1\right)\right)\right\}_{i=1,2,3,4}; \hat{\theta}_2\right)\right. \tag{5}$$

where $\hat{\theta}_1$ and $\hat{\theta}_2$ are the parameters of well-trained $\mathcal{F}_B$ and $\mathcal{F}_S$.

After the above self-SL training process is completed, the proposed BockNet incorporates a sliding dual-window reconstruction model. The appearance of the guard window diminishes the interference of anomalies, making BockNet act as a very efficient background reconstruction network. Furthermore, the reconstructed anomalous targets will approximate the surrounding background regardless of their size and cannot successfully restore the original anomalous properties. Accordingly, the reconstructed HSI is regarded as a background HSI, and anomalies have higher reconstruction errors compared to

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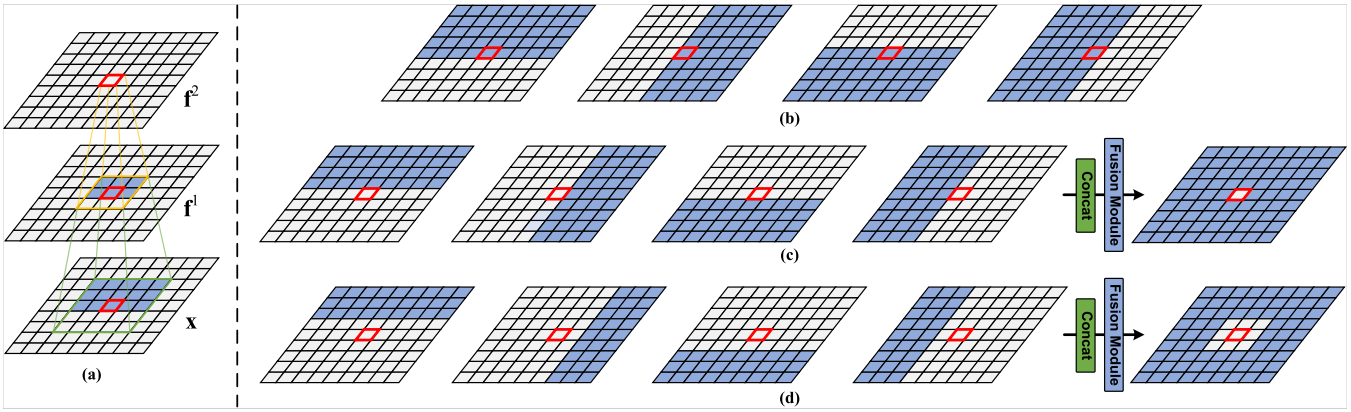


Fig. 5. Let us take the receptive field as an example to explain the principle of BockNet setting the blind area (guard window) on the input HSI. The red rectangles indicate central pixels, the light squares within the original receptive field areas represent the blind areas, and the remaining blue squares need to be calculated together to get the value of the central pixel. (a) Example of a two-layer convolution operation with an offset process demonstrates that the actual receptive field area (blue squares) of the central pixel of the feature map $\mathbf{f}^2$ on the feature map $\mathbf{f}^1$ and the input $\mathbf{X}$ grows along a specific direction (upward). (b) Actual receptive field extends in four directions after the rotation operation corresponding to the center pixel before performing the shift operation, where the center pixel is included. (c) After the shift operation (with a size of 1) and fusion, the final receptive field associated with the central pixel is obtained, and its center is the blind-spot (i.e., no guard window). (d) After the shift operation (with a size of 2) and fusion, the final receptive field related to the central pixel is obtained, and its 3×3 central block is the blind area (with a guard window of size 3×3). The pixels lying in the central block of the receptive field are invisible to our BockNet, which therefore predicts/reconstructs the central pixel using only the surrounding pixels. Our blind-block is considered as the guard window.

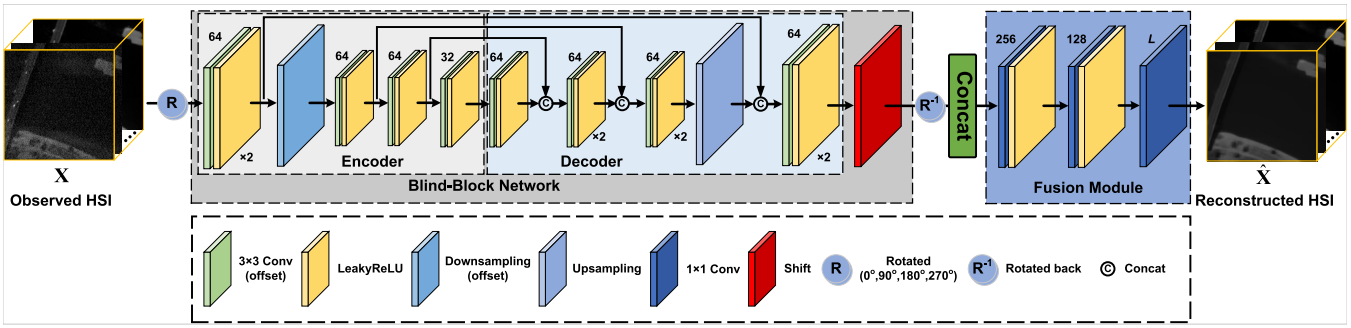


Fig. 6. Details of the proposed overall network structure (the numbers above Conv, such as 32, 64, and 128, represent the number of output channels; the mark offset indicates that the input feature map requires to be offset in advance before performing the convolution or pooling operation).

backgrounds. Finally, the reconstruction error given in the following equation is taken as the HAD result:

$$D_{i,j} = \|\mathbf{x}_{i,j} - \hat{\mathbf{x}}_{i,j}\|_2 \tag{6}$$

where $\mathbf{x}_{i,j} \in \mathbb{R}^{L\times 1}$ and $\hat{\mathbf{x}}_{i,j} \in \mathbb{R}^{L\times 1}$ denote the pixels at location (i, j) of $\mathbf{X}$ and $\hat{\mathbf{X}}$, respectively, $D_{i,j}$ stands for the anomaly scores of the corresponding location pixels and constitutes the final HAD map $\mathbf{D} \in \mathbb{R}^{R\times R}$. The key procedures of the proposed BockNet are outlined in Algorithm 1.

III. Experimental Results

In this section, the superiority of the proposed BockNet for HAD is corroborated qualitatively and quantitatively through comprehensive experiments.

A. Datasets

1) Airport Dataset: It was collected by airborne visible infrared imaging spectrometer (AVIRIS) covering Gulfport city [71], and the anomalous objects in this HSI are three aircraft with different sizes and shapes.

2) Beach Dataset: It was derived from the reflective optics system imaging spectrometer (ROSIS-03) sensor over a beach area of Pavia [71], in which the anomalies mainly consist of some vehicles on the bridge.

3) Urban Dataset: It was captured by the AVIRIS, which covered an urban area in Gainesville city [71]. There are several small-sized anomalous targets embedded in the background, most of which are ships.

4) San Diego Dataset: It was also acquired from the AVIRIS covering the San Diego airport scene [38], and three aircraft of larger size are considered to be anomalies in this dataset.

Notably, the above four real images contain both small- and large-sized anomalous targets, and more detailed characteristics of these images are listed in Table I. Additionally, Fig. 7 exhibits these datasets' pseudo-color images and ground-truth maps of targets.

B. Comparison Algorithms and Evaluation Metrics

1) Comparison Algorithms: To evaluate the performance of the proposed BockNet, ten well-established detectors selected for comparison embrace five methods based on sliding

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Algorithm 1 Proposed BockNet for HAD

Input: Observed HSI $\mathbf{X} \in \mathbb{R}^{R\times R\times L}$; Parameter: width of guard window B_s.

Output: Final detection result $\mathbf{D} \in \mathbb{R}^{R\times R}$.

Stage 1: Training Phase;

Obtain the rotated HSIs $\mathbf{X}_{Ri} \in \mathbb{R}^{R\times R\times L}$, $i = 1, 2, 3, 4$ using the rotation operation by Eq. (1);

Acquire the output $\mathcal{F}_B(\mathbf{X}_{Ri}; \theta_1)$ of the network $\mathcal{F}_B$ with a guard window of size $B_s \times B_s$, parameterized by θ_1, and $\mathcal{F}_B$ takes $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$ as input;

Obtain the final reconstructed image $\widetilde{\mathbf{X}} \in \mathbb{R}^{R\times R\times L}$ by performing rotation, concatenation, and fusion operations using Eq. (2);

Train BockNet by minimizing Eq. (3);

Gain the optimal BockNet, parameterized by $\hat{\theta}$ (composed of $\hat{\theta}_1$ and $\hat{\theta}_2$);

Stage 2: Detection phase;

Obtain the rotated HSIs $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$, using the rotation operation, by Eq. (1);

Acquire the reconstructed HSI $\widehat{\mathbf{X}} \in \mathbb{R}^{R\times R\times L}$ by feeding $\mathbf{X}_{Ri}, i = 1, 2, 3, 4$ into the well-trained BockNet using Eq. (5);

Calculate the degree of anomaly $D_{i,j}$ of each pixel in $\mathbf{X}$ by Eq. (6);

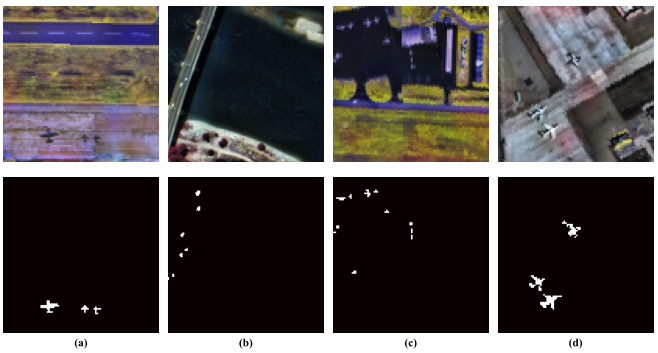


Fig. 7. Pseudo-color images and ground-truth maps of the four considered HSI datasets. (a) Airport. (b) Beach. (c) Urban. (d) San Diego.

TABLE I
DETAILS OF THE EXPERIMENTAL DATASETS

Data set	Location	Sensor	Image size	Resolution
Airport	Gulfport	AVIRIS	$100 \times 100 \times 191$	3.4m
Beach	Pavia	ROSIS-03	$150 \times 150 \times 102$	1.3m
Urban	Gainesville	AVIRIS	$100 \times 100 \times 191$	3.5m
San Diego	San Diego	AVIRIS	$100 \times 100 \times 189$	3.5m

dual-window model (i.e., LRX [10], SRX_Local [11], CRD [31], CRDBPSW [33],[1] and 2S-GLRT [13][2]) and five global-based methods (i.e., GRX [9], FrFE [12],[3] GAED [49],[4] RGAE [48],[5] and Auto-AD [50][6]). Among the above methods, based on advanced DL technology, GAED, RGAE, and Auto-AD methods aim to strengthen the ability of models to reconstruct the background and all adopt the residual spectral error between the original HSI and the reconstructed one to measure the abnormal level of each pixel. This remains consistent with the motivation of our proposed BockNet. Thus, the differences between the detection effects of these four methods require to be focused on to demonstrate the effectiveness of the blind-block model with a guard window.

2) Evaluation Metrics: To assess the detection performance of the different compared approaches, some well-known metrics are adopted, that is, the receiver-operating characteristic (ROC) curve, the area under the ROC curve (AUC) [72], and the statistical separability map. The ROC curve indicates the relationship between the probability of detection and false alarm, reflecting the detection accuracy of the method in a holistic mode. The corresponding AUC value can quantitatively evaluate the detection accuracy. An ideal detector would take along a ROC curve in close proximity to the upper left corner, with an AUC value near 1. Moreover, the separability boxplot gives the statistical range of the abnormal response values at the pixel locations of the background and anomaly categories, which are distributed in two separate boxes (blue for the background and red for the anomaly), each with a statistical range from 10% to 90%. Obviously, if the detector shows a predominant capability to suppress backgrounds and extrude anomalies, the blue box will be at the very bottom and narrow in width, with a large gap from the red box.

C. Detection Performance for Different Methods

Figs. 8–11 present the HAD results obtained by the 11 considered detectors corresponding to four benchmark datasets, where the first in each figure is the ground truth of anomalies to be detected, and the rest are the result maps for different methods. Furthermore, Figs. 12 and 13 show the ROC curves and separability boxplots of these detectors, whose AUC and false alarm rate (FAR) scores for the fixed detection rate of 1 are listed in Tables II and III, respectively. Note that the best results are marked in boldface (the same as below).

Across the board, Figs. 8–11 reveal that the proposed BockNet achieves a satisfactory tradeoff in anomaly detectability and background suppressibility, and its corresponding detection maps are closest to the ground truth. Considering the Airport dataset as an example, LRX, 2S-GLRT, GRX, and FrFE do not work satisfactorily, because the shapes of two small-sized aircraft are barely visible in the results. Although CRD, GAED, RGAE, and Auto-AD perform relatively well in highlighting anomalous targets, they retain some background structural information and suffer from many false alarms. SRX_Local and CRDBPSW can adequately restrain the

[1]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-CRDBPSW

[2]https://github.com/zephyrhours/Hyperspectral-Anomaly-Detection-2S-GLRT

[3]https://github.com/xudongzhao461/Hyperspectral-Anomaly-Detection-by-Fractional-Fourier-Entropy

[4]https://github.com/pei-xiang/GAED

[5]https://github.com/FGH00292/Hyperspectral-anomaly-detection-with-RGAE

[6]https://github.com/RSIDEA-WHU2020/Auto-AD

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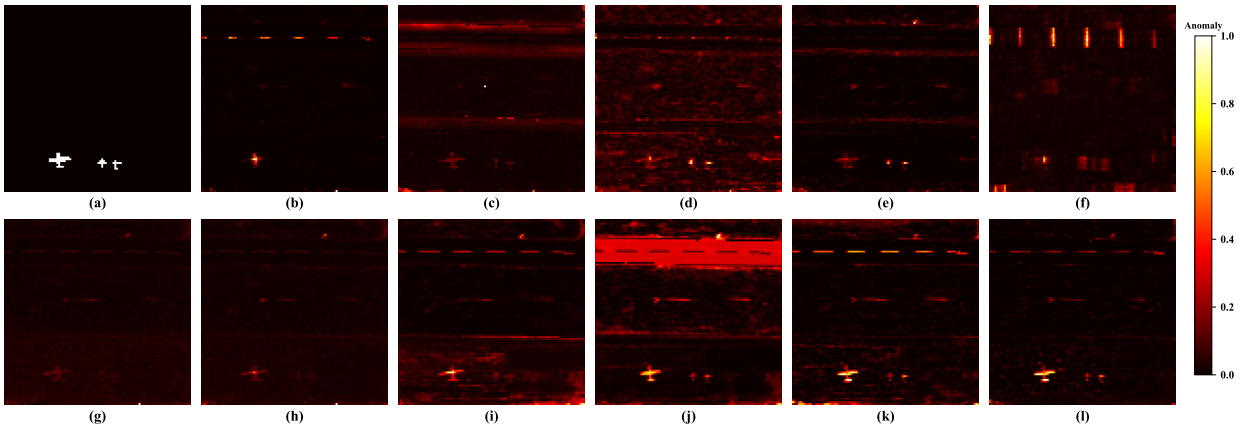


Fig. 8. HAD colored maps obtained by different algorithms for the Airport dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) GAED. (j) RGAE. (k) Auto-AD. (l) BockNet.

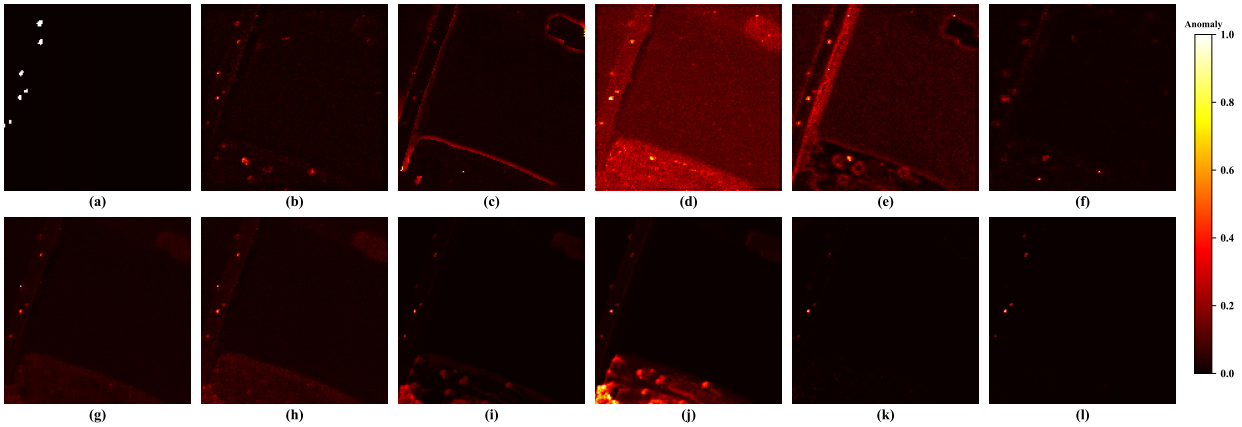


Fig. 9. HAD colored maps obtained by different algorithms for the Beach dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) GAED. (j) RGAE. (k) Auto-AD. (l) BockNet.

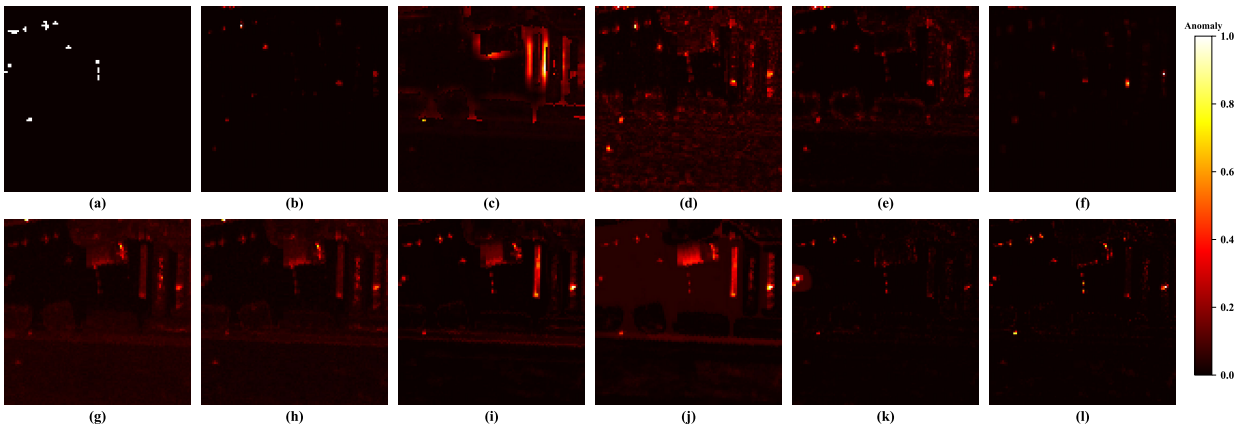


Fig. 10. HAD colored maps obtained by different algorithms for the Urban dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) GAED. (j) RGAE. (k) Auto-AD. (l) BockNet.

background, but cannot effectively highlight the targets. Conversely, our proposed BockNet avoids the above-mentioned problems and can accurately identify the three aircraft with different sizes with high response values and low false alarms.

Another example is the San Diego dataset, where LRX and 2S-GLRT fail to detect anomalies. SRX_Local, CRD, CRDBPSW, GRX, and FrFE can locate the position of the aircraft, but suffer from varying degrees of background interference, making the distinction between background and anomaly not evident. GAED, RGAE, and Auto-AD can suppress background interference to a certain extent. Still, our BockNet is more capable of preserving detailed information about the aircraft, especially that located above, compared to these DL-based methods, while yielding a lower FAR.

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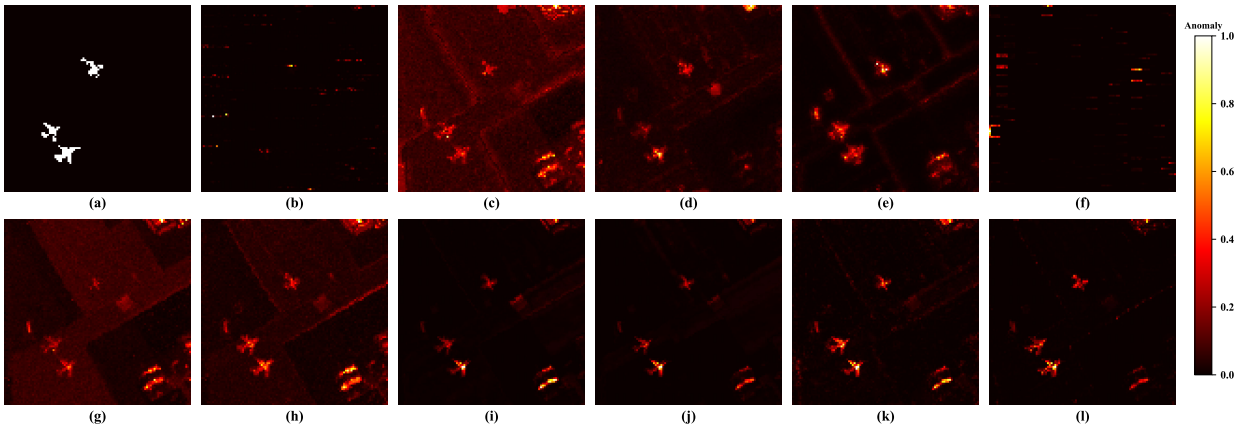


Fig. 11. HAD colored maps obtained by different algorithms for the San Diego dataset. (a) Ground truth. (b) LRX. (c) SRX_Local. (d) CRD. (e) CRDBPSW. (f) 2S-GLRT. (g) GRX. (h) FrFE. (i) GAED. (j) RGAE. (k) Auto-AD. (l) BockNet.

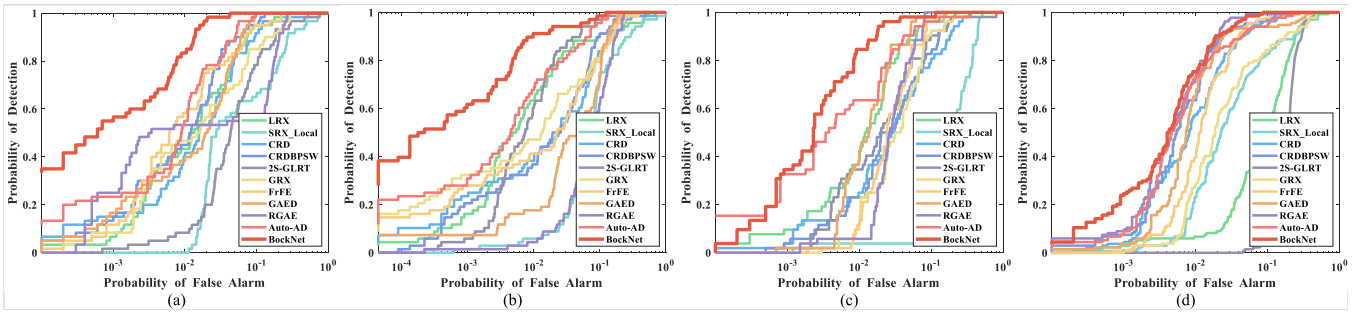


Fig. 12. ROC curves obtained by different detectors for the four considered datasets. (a) Airport. (b) Beach. (c) Urban. (d) San Diego.

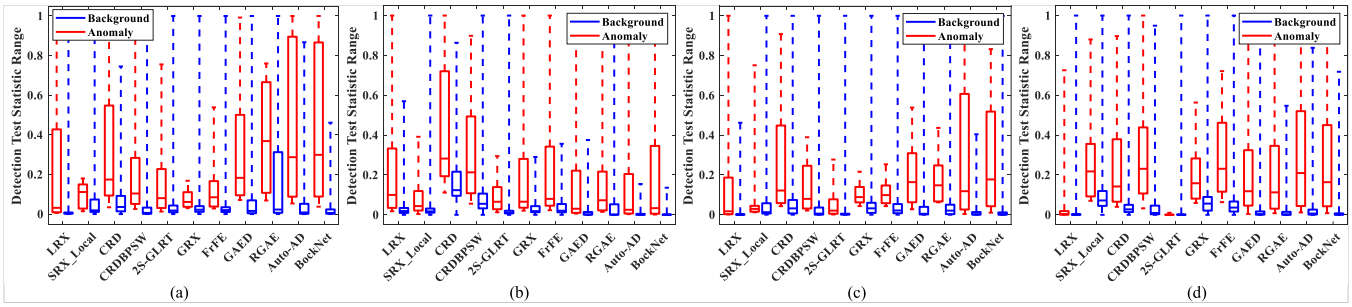


Fig. 13. Separability maps obtained by different detectors on the four considered datasets. (a) Airport. (b) Beach. (c) Urban. (d) San Diego.

Similar results are presented on the other two datasets, and these visualizations show that BockNet has a better capacity for separating targets from backgrounds. The main reason is that the proposed BockNet adopts a blind-block network with the guard window in a self-supervised manner, indicating promising performance in enhancing anomalous characteristics and suppressing background interference.

Besides, other evaluation metrics were employed to evaluate the performance of different algorithms both qualitatively and quantitatively. In Fig. 12, overall, the red curve of BockNet is mostly located above the other colored curves and generally outperforms the other methods. Unsurprisingly, BockNet achieves the highest AUC scores compared with the other competing methods. To further demonstrate BockNet's strong separability for anomalous targets and suppressibility for backgrounds, Fig. 13 draws separability maps of all methods regarding anomalies and backgrounds. On these four datasets, the red anomaly boxes of BockNet are commonly not located at the highest position, but its blue background boxes are narrow enough and at a rock-bottom level, hinting that BockNet is capable of restraining the background. At the same time, on the three datasets (except for the Beach dataset), BockNet can ensure a more evident gap between the two boxes and swimmingly extract anomalies from the suppressed background. Obviously, this is in agreement with the above-mentioned visual analysis results.

Given all of that, from the HAD maps, the proposed BockNet exhibits impressive visual results. In addition, the comparison and analysis of the ROC curves, AUC scores, and separability boxplots for different detectors reveal that BockNet obtains the highest average AUC of 0.9933, which is 0.0115 larger than that of the second-best Auto-AD method,

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TABLE II
AUC VALUES OF THE 11 CONSIDERED DETECTORS FOR DIFFERENT DATASETS

Data Set	Detectors and AUC values										
	LRX	SRX_Local	CRD	CRDBPSW	2S-GLRT	GRX	FrFE	GAED	RGAE	Auto-AD	BockNet
Airport	0.9694	0.8847	0.9599	0.9791	0.9183	0.9526	0.9744	0.9746	0.9184	0.9839	**0.9957**
Beach	0.9590	0.8795	0.9407	0.9552	0.9803	0.9538	0.9498	0.9394	0.9094	0.9820	**0.9931**
Urban	0.9769	0.7615	0.9393	0.9522	0.9485	0.9513	0.9645	0.9797	0.9665	0.9787	**0.9929**
San Diego	0.8795	0.9317	0.9809	0.9872	0.7973	0.9403	0.9815	0.9743	0.9905	0.9824	**0.9913**
Average	0.9462	0.8644	0.9552	0.9684	0.9111	0.9495	0.9676	0.9670	0.9462	0.9818	**0.9933**

TABLE III
FAR VALUES OF THE 11 CONSIDERED DETECTORS FOR DIFFERENT DATASETS

Data Set	Detectors and FAR values										
	LRX	SRX_Local	CRD	CRDBPSW	2S-GLRT	GRX	FrFE	GAED	RGAE	Auto-AD	BockNet
Airport	0.2748	0.7339	0.5244	0.1298	0.7562	0.2911	0.2182	0.0959	0.2648	0.0966	**0.0422**
Beach	0.5420	0.9980	0.6913	0.4913	0.3616	0.3646	0.3861	0.2032	0.2280	0.1391	**0.1240**
Urban	0.1548	0.5141	0.3342	0.2147	0.8114	0.2398	0.1656	0.1529	**0.0816**	0.3542	0.1150
San Diego	0.7397	0.5293	0.1994	0.1585	0.5384	0.4738	0.1158	0.4239	0.2308	0.2813	**0.0851**
Average	0.4279	0.6938	0.4373	0.2486	0.6169	0.3424	0.2214	0.2190	0.2013	0.2178	**0.0916**

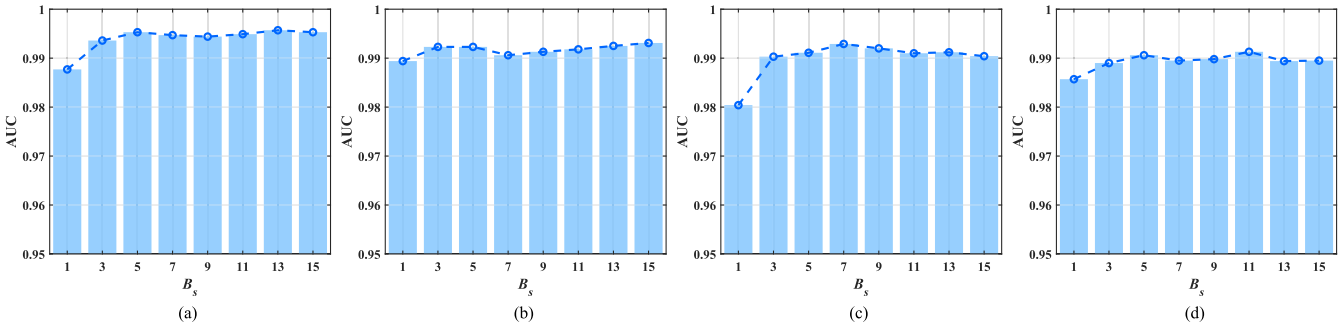


Fig. 14. AUC values of the proposed BockNet with guard windows of different widths (B_s) on the four considered datasets. (a) Airport. (b) Beach. (c) Urban. (d) San Diego.

as reported in Table II. Meanwhile, the FAR of BockNet is normally maintained at low levels, and BockNet obtains the lowest average FAR of 0.0916, as shown in Table III, further achieving a satisfactory balance between anomaly detectability and background suppressibility.

D. Parameter Analysis

The proposed BockNet involves a parameter B_s, the width of the square blind-block (i.e., the width of the guard window). It is worth noting that the feasible value of B_s needs to be an odd number greater than 1. If B_s is equal to 1, then BockNet degenerates to a blind-spot network without a guard window. Among them, the value of B_s for all the datasets is set to {1, 3, 5, 7, 9, 11, 13, 15}, and the AUC scores under different parameter settings are illustrated in Fig. 14.

It is found from Fig. 14 that the detection performance of BockNet with a guard window of different sizes (greater than 1) does not display significant fluctuations on each dataset, manifesting its stability for detecting anomalous targets of different spatial sizes. Concretely, for datasets with large-sized targets (such as Airport and San Diego), B_s should be chosen as a larger value to further promote the performance of BockNet. For the four datasets containing targets of different sizes, the parameter B_s is fixed to 13, 15, 7, and 11, respectively, as the optimal setting.

E. Model Verification

1) Ablation Study: In this section, the contribution of each component in the proposed BockNet is evaluated on all datasets. We use the traditional USL-based reconstruction model as the baseline, which employs the same framework as that of BockNet, except for the fact that the regular convolution and pooling layers without offset operations are adopted, and the original HSI (rather than four rotated HSIs) is fed directly into the network. BockNet offsets the feature map of each convolution and pooling layer and then acquires a receptive field extending around but not including the central blind area (blind-spot or blind-block) through shift and fusion operations. Finally, the corresponding center pixels are reconstructed using pixels outside the blind area in a self-SL way. If the size of the

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TABLE IV
AUC VALUES AFTER THE ABLATION STUDY FOR DIFFERENT DATASETS

Data Set	Detectors and AUC values			
	USL-based	BSNet	BockNet (ℓ_2)	BockNet (ℓ_1)
Airport	0.9571	0.9877	0.9957	**0.9981**
Beach	0.9829	0.9894	0.9931	**0.9950**
Urban	0.9789	0.9804	**0.9929**	0.9913
San Diego	0.9852	0.9857	**0.9913**	0.9905

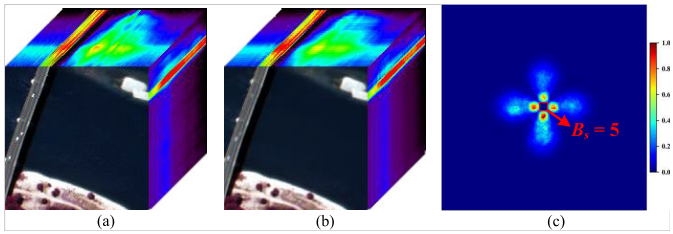


Fig. 15. Visualization of (a) original image, (b) reconstructed image obtained by BockNet, and (c) effective receptive field [73] corresponding to the center pixel when $B_s = 5$ on the beach dataset.

guard window is set to 1×1, the blind area of the receptive field is a blind-spot, and BockNet degenerates into a blind-spot network (denoted as BSNet). In contrast, the receptive field of BockNet contains the guard window (the blind area is a blind-block) to form a dual-window model. Meanwhile, we also compared the impact of different types (i.e., ℓ_1 and ℓ_2) of reconstruction errors on the model detection accuracy. The specific AUC comparisons on these datasets are summarized in Table IV.

As seen from Table IV, BSNet has certain advantages over the USL-based method, but when reconstructing pixels inside large-sized anomalous targets, both the USL-based network and self-SL-based BSNet (spatial–spectral joint learning-based models) are susceptible to interference from neighboring anomalous pixels, which results in model overfitting anomalies and limits detection performance. While BockNet (coupled with the sliding dual-window model) avoids the aforementioned problems, the guard window can prevent the potential contamination of anomalies to reconstruct an HSI with a purer background. Thus, the ability of BockNet to identify anomalies from HSIs is significantly promoted, leading to higher AUC results. In addition, the reconstruction error based on ℓ_1 and ℓ_2 as the degree of anomaly of each pixel has a comparable impact on the model to separate anomalous samples from the original HSI, which side-by-side reflects the robustness of BockNet in the manner of measuring anomalies.

2) Feature Visualization: Fig. 15 visualizes the original image, the reconstructed image obtained by BockNet, and the effective receptive field corresponding to the central pixel when $B_s = 5$ on the Beach dataset. For the visualization of the receptive fields, we calculated the gradient response values associated with the central pixel. Note that the brighter color indicates a greater contribution to reconstructing the central pixel. As can be viewed in Fig. 15(a) and (b), the background in the reconstruction image obtained by BockNet is adequately reconstructed, but most of the anomalous targets (vehicles) are not. The network erases the anomalous attributes while endowing the targets with the surrounding background characteristics, which leads to the background with small reconstruction errors and the anomalies with large reconstruction errors. To trace the reason, our BockNet creates a guard window for reconstructing each pixel, and the guard window is achieved by setting a blind-block (the network cannot see this information) at the center of the receptive field, while the receptive field outside the guard window is deemed as the outer window, which is visible to the network for reconstructing the center pixel. When the central pixel is anomalous, the sparsely distributed and low-proportion anomaly targets typically lie in the guard window. Therefore, the network will not see these anomalous properties, and BockNet has difficulty reconstructing the anomalous pixels. Meanwhile, since the background is generally a large-scale homogeneous region, which tends to be located within the outer window, the reconstructed anomalous pixels seem to converge to the background distribution of the outer window, while the background pixels can be reconstructed sufficiently. Fig. 15(c) displays the actual effective receptive field corresponding to the center pixel when the size B_s of the guard window of BockNet is 5. We can find that the gradient response value of the central block in the receptive field is 0, indicating that it is not involved in the reconstruction of the central pixel and exerts the role of the guard window. The receptive field regions outside the central block, that is, the outer window, present varying degrees of gradient values, reflecting that the corresponding outer window pixels do their part to achieve the goal of reconstructing the central pixel.

F. Performance Analysis Between the Target Size and the Guard Window Size

To investigate the relationship between the target size and the guard window size, we regularly embed 2×2 anomalous targets on a background subimage, and their sizes are set from 1×1 to 11×11. Among them, the original background image and anomaly spectrum (building) were selected from the Salinas scene [74]. Meanwhile, we refer to the anomalous target implantation approach of [75], which follows a linear mixing model [76], [77], [78]. Then, the abundance of all targets and the width of guard window B_s were set to 0.3 and $\{1, 3, 5, 7, 9, 11\}$, respectively. Next, the AUC values achieved by the proposed BockNet on the 11 synthetic datasets (using different guard window sizes) are recorded in Table V. We reiterate that when B_s is set to 1, BockNet degenerates to a blind-spot network without the guard window (same as BSNet in Section III-E).

We found that BSNet suffered from varying degrees of performance degradation on HSIs with a target size larger than 5×5, and the larger the target size, the lower the corresponding AUC score. Moreover, when B_s is small (e.g., 3 and 5), BockNet appears to exhibit an analogous behavior on HSIs with larger target sizes. However, with the increase of B_s, the applicability of BockNet turns wider, and, in particular, BockNet always enables optimal performance on HSIs containing synthetic targets of various sizes when the width of the guard window is 11. This is because BockNet employs a sliding dual-window-based reconstruction model, where the inner window

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TABLE V
AUC VALUES OF THE PROPOSED BOCKNET WITH DIFFERENT PARAMETERS ON THE SYNTHETIC DATASET

Width of guard window (B_s)	Target size										
	1×1	2×2	3×3	4×4	5×5	6×6	7×7	8×8	9×9	10×10	11×11
1	1.0000	0.9999	0.9997	0.9903	0.9846	0.9800	0.9642	0.9576	0.9413	0.9410	0.9251
3	1.0000	1.0000	1.0000	0.9998	0.9989	0.9947	0.9903	0.9790	0.9622	0.9418	0.9487
5	1.0000	1.0000	1.0000	1.0000	0.9999	0.9980	0.9937	0.9921	0.9875	0.9815	0.9672
7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9983	0.9940	0.9927	0.9913	0.9757
9	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9994	0.9977	0.9951	0.9940
11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9994	0.9989	0.9959

TABLE VI
RUNNING TIMES (IN SECONDS) OF THE 11 CONSIDERED DETECTORS

Detector		Data set and running times			
		Airport	Beach	Urban	San Diego
LRX		27.3184	17.4160	20.9789	24.3916
SRX_Local		3.4674	2.9611	2.8974	2.8699
CRD		3.8910	2.9411	2.2356	5.1533
CRDBPSW		13.9485	26.7408	5.2544	57.0158
2S-GLRT		53.5927	22.7698	22.7819	21.1771
GRX		0.0589	0.0486	0.0493	0.0519
FrFE		10.4997	11.7432	9.5050	10.5576
GAED	Train	41.3743	78.5159	39.3869	39.2719
	Test	0.0526	0.0692	0.0440	0.0403
RGAE	Train	191.0665	292.1609	188.8200	187.1919
	Test	**0.0243**	0.0307	**0.0207**	0.0191
Auto-AD	Train	10.2913	24.5171	10.1628	8.2967
	Test	0.0266	**0.0299**	0.0209	**0.0171**
BockNet	Train	75.4866	107.6639	64.5493	58.0618
	Test	0.9538	0.9598	0.9670	0.9494

(guard window) lessens the interference of anomalies with regards to the outer window (the receptive field outside the guard window) pixels, boosting the model's reconstruction ability for background samples and weakening the model's representation ability for anomalous samples to obtain a purer background. Especially for large-sized targets, BockNet can make full use of the self-SL-based blind-spot framework for HAD purposes.

G. Comparison of Running Times

The running times of different methods on all datasets are listed in Table VI to evaluate the computational burden of these models. Except for Auto-AD and BockNet, which were implemented based on the PyTorch 1.12.1 framework, the remaining methods were implemented using MATLAB 2019b. Compared to the traditional detectors using some tricks (such as fractional Fourier transform and sliding dual-window), GRX obtains the shortest computation time by virtue of its clean design principle, but its detection effect is relatively inferior. The four DL-based algorithms in the experiments require additional steps to train the network parameters, among which BockNet consumes less training time than RGAE, and Auto-AD achieves the shortest training time since it does not involve pre- and postprocessing procedures. Moreover, these four algorithms exhibit comparable inference times on different datasets, because they employ the same procedure of detecting anomalies (the network is expected to be a superior background reconstruction model, and the MSE-based reconstruction error is adopted as the final anomaly score). Yet, the proposed BockNet also requires rotating the original HSI and feeding all four rotated HSIs into the network, so it takes a little extra time. Nevertheless, our BockNet achieves promising performance, and its demonstrated competitiveness is unquestionable.

IV. CONCLUSION

This article creatively embeds a sliding dual-window reconstruction model in a deep network for HAD. To be specific, we built a simplified U-Net—with the receptive field growing only upward along the center row—which allows the spatial–spectral characteristics of the HSI to be exploited adequately. In addition, four rotated HSIs are fed into the network, and the produced feature maps are shifted and fused to yield the final receptive field, which extends in each direction along the central blind-block, treated as a guard window (the network cannot see this information). Eventually, our BockNet divides its receptive field into the dual-window and reconstructs each pixel using only the outer window (the receptive field outside the guard window) pixels. Hence, there is a reduced interference of anomalous samples to the reconstructed HSI, enabling the proposed BockNet to emerge as a promising background reconstruction model. Experiments demonstrate the leading and comprehensive performance of our BockNet on HSIs containing anomalous targets of different sizes. However, the proposed BockNet is trained in a fully self-supervised manner on an unlabeled HSI and performs anomaly detection on the same data. Due to the significant differences in HSIs acquired by different sensors under different imaging conditions, the network is ineffective in applying directly to different images, limiting the generalization of BockNet. Therefore, in the future, we will design new frameworks with strong generalizability and transferability.

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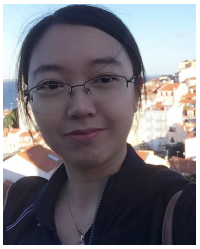
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